

PROBLEM SET 1

Due date: March 21, 2013 at the recitation session

1. Consider the following game

		Pl 2		
		L	M	R
Pl 1	S_1	2, 2	3, 0	4, 0
	S_2	3, 3	2, 0	1, 0
	S_3	1, 3	5, 5	0, 2
	S_4	1, 1	1, 1	2, 3

Figure 1

- (a) Iteratively eliminate all strictly dominated strategies. Which strategies survive?
 - (b) What are the rationalizable strategies?
2. Consider a game between an employer and a worker, in which the employer provides the capital K and the worker provides the labor L to produce $f(K, L) = \sqrt{KL}$. The per-unit costs of capital and the labor for the employer and the worker are $r > \frac{1}{4}$ and $c > \frac{1}{4}$, respectively. The firm and the employee simultaneously choose K and L respectively and share the corresponding output equally. The worker cannot invest more than $\bar{L} > 0$.

 - (a) Write down the payoff functions for the firm $U_f(K, L)$ and the employee $U_e(K, L)$.
 - (b) Find all the rationalizable strategies.
3. Consider a second-price sealed bid auction game with two bidders $I = \{1, 2\}$. The seller has one indivisible object. The valuation for each bidder i is given by v_i with $v_1 \leq v_2$. These valuations are common knowledge. The bidders simultaneously submit bids $s \in [0, \infty)$. The bidder with the highest bid wins the object and pays the *second highest bid*. If the two players submit the same bid, each one of them obtains the object with probability $\frac{1}{2}$.

 - (a) Write down the payoff function $u_i(s_i, s_j)$.
 - (b) Does any of the players have strictly dominated strategies? If so, what are they?
 - (c) Describe the set of rationalizable strategies. Can you eliminate any strategies on the bases of them not being rationalizable?
 - (d) s_i is a *weakly* dominated strategy if there exists another strategy s'_i such that $u_i(s'_i, s_j) \geq u_i(s_i, s_j)$. Does player i have weakly dominated strategies? If so, what are they?
 - (e) Consider now an iterative elimination of weakly dominated strategies. Describe one strategy pair that survives iterated elimination of weakly dominated strategies? Is this pair unique? What is the winning probability and payoff of each player i using this solution concept? Are these unique?

4. Consider the following game:

		A P 2				B P 2	
		<i>L</i>	<i>R</i>			<i>L</i>	<i>R</i>
P 1	<i>U</i>	9	0	P 1	<i>U</i>	0	9
	<i>D</i>	0	0		<i>D</i>	9	0

		C P 2				D P 2	
		<i>L</i>	<i>R</i>			<i>L</i>	<i>R</i>
P 1	<i>U</i>	0	0	P 1	<i>U</i>	6	0
	<i>D</i>	0	9		<i>D</i>	0	6

Player 1 chooses rows, player 2 chooses columns and player 3 chooses matrices. Player 3's payoff are given in the figure. Show that action D is not a best response to any mixed strategy of players 1 and 2, but that D is not dominated.